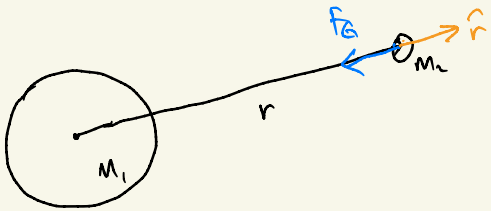
Resnick Ch 13: Gravitation

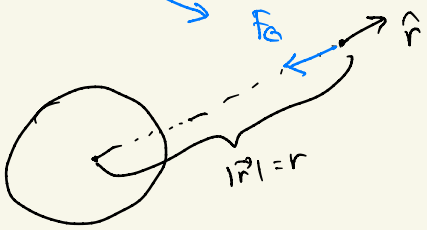
- When you compute potential E, (+) or (-) what does the sign mean?

(+) repulsive
(-) attractive

mgh: relative potential (exception)

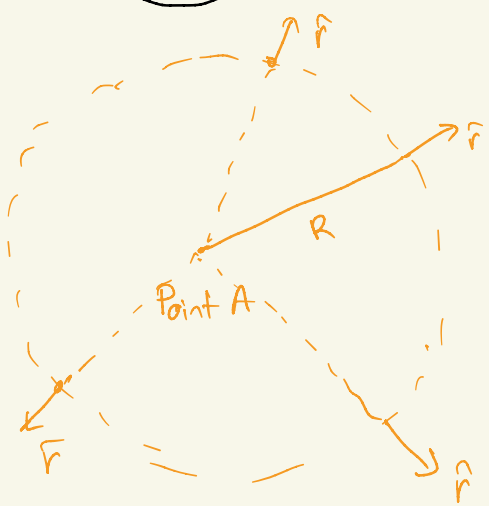


$$\vec{F} = -\frac{GM_1m_2}{r^2}\hat{r}$$

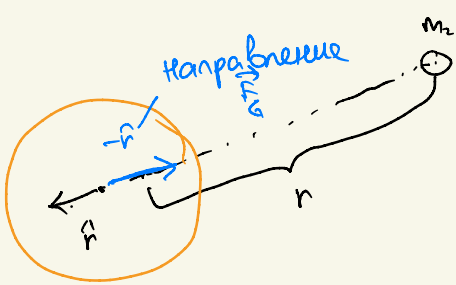


$$\vec{r} = r\hat{r} = |\vec{r}|\hat{r}$$

unit vector

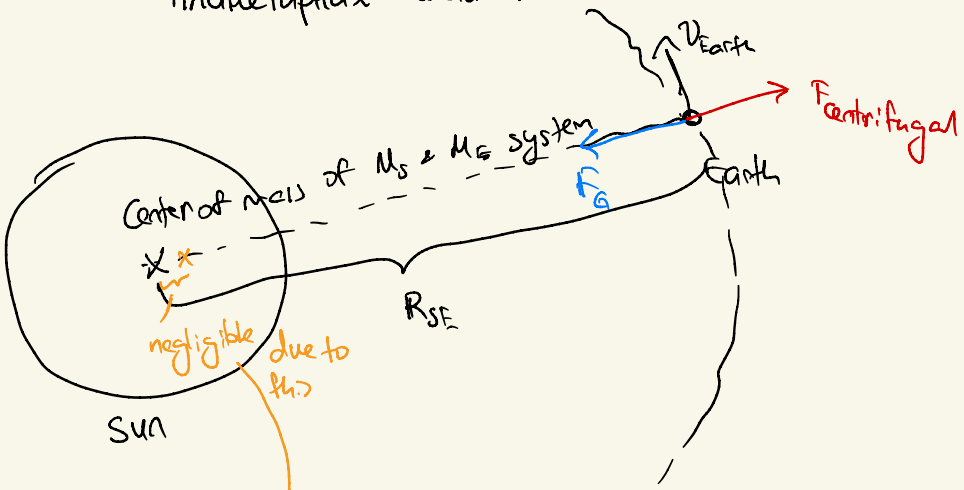


$\hat{r}$ is a unit vector from source pointing radially outward



$$\vec{F}_{net} = \sum_{i=1}^{N} \vec{F}_i$$

Планетарная система:



Here, we've assumed $M_S \gg M_E$

$$-\frac{GM_SM_E}{R^2} + \frac{M_E v_E^2}{R} = 0$$

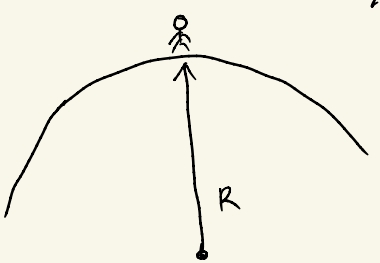
$$v_E = \sqrt{\frac{GM_S}{R}}$$

Earth:

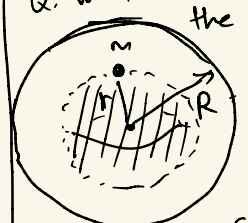
$$mg = \frac{GMm}{R^2}$$

масса земли

$$g = \frac{GM}{R^2}$$



Q: What will be force of gravity within the planet?



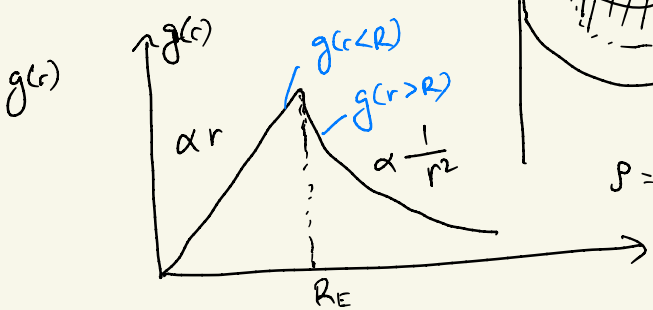
$$F_G = -\frac{GM_{inside}M}{r^2}$$

$$mg = \frac{GM_{in}M}{r^2}$$

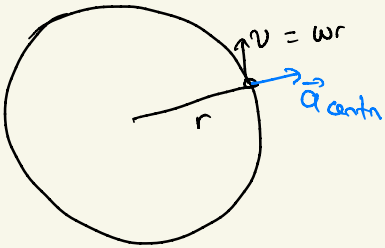
$$g(r<R) = \frac{GM_{in}}{r^2}$$

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3}, \quad M_{in} = \rho \cdot \frac{4}{3}\pi r^3$$

$$g(r) = \frac{G \cdot \rho \cdot \frac{4}{3}\pi r^3}{r^2} = G\frac{M \cdot \frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3 r^2} = \frac{GM_r}{R^3}$$



Uniform Circular Motion:



$$\omega = \frac{2\pi}{T} = 2\pi f$$

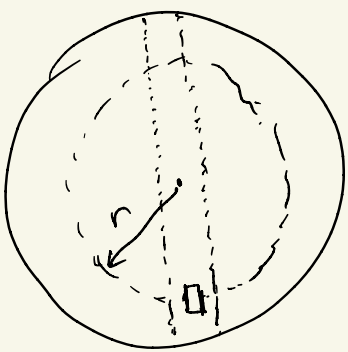
T: Period of revolution (seconds) ← unit

f: frequency ($Hz = 1s^{-1}$)

$$a = \omega^2 r = \frac{v^2}{r}$$

$$= (2\pi f)^2 r = \frac{4\pi^2}{T^2} r$$

Ex: Tunnel through Earth's center

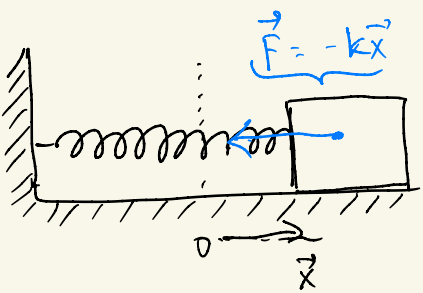


$$mg = \frac{GM_{in}m}{r^2}$$

$$g(r) = \frac{GM_{in}}{r^2} = \frac{GMr}{R^3}$$

$$F_g = mg = -\frac{GMm}{R^3} r\hat{r}$$

This similar to mass-spring system



$$F = ma = m\frac{dv}{dt} = m\frac{d^2x}{dt^2}$$

$$-kx = m\frac{d^2x}{dt^2} = m\ddot{x}$$

($\ddot{x}$: 2nd order differential)

$$\boxed{m\ddot{x} + kx = 0}$$

In general, $\frac{d^2x}{dt^2} + \omega^2 x = 0$ is the eqn of harmonic oscillation

$\ddot{x} + \frac{k}{m} x = 0$

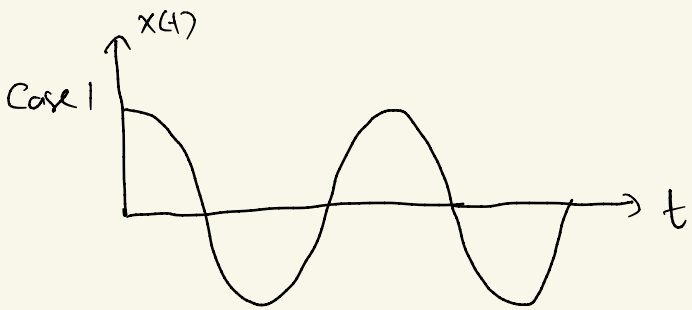
Частота колебания: $\omega = \sqrt{\frac{k}{m}}$

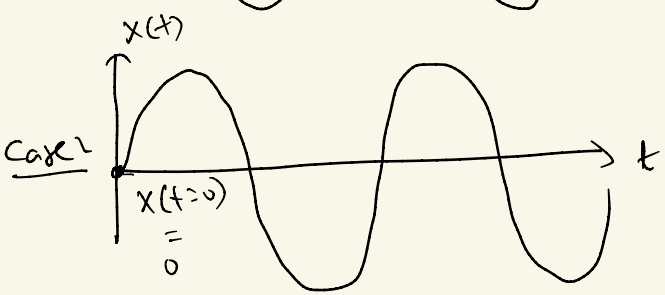
solution of this 2nd order homogeneous differential eqn

$$x(t) = A\cos\omega t + B\sin\omega t$$

is the general solution of all $\ddot{x} + \omega^2 x = 0$

- A, B can be found from initial conditions

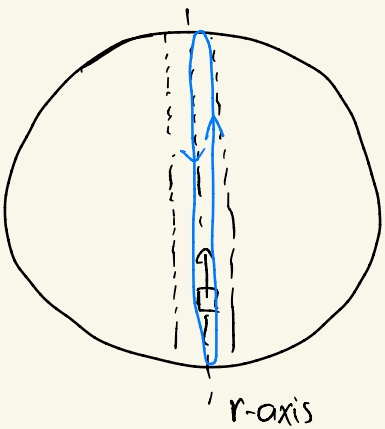




$x(t=0) = A\cos 0 + B\sin 0$

$= A$

$= 0$



$$F_G = ma$$

$$-\frac{GMr}{R^3} = m\frac{d^2r}{dt^2}$$

$$\ddot{r} + \frac{GM}{mR^3}r = 0$$

we know the solution for this

$$\ddot{r} + \omega^2 r = 0$$

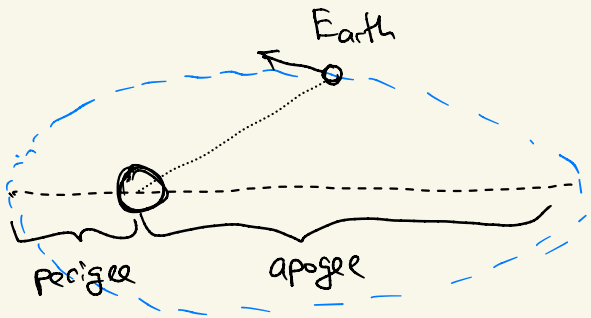
$$\omega = \sqrt{\frac{GM}{mR^3}}$$

$$r(t) = A\cos\left(\sqrt{\frac{GM}{mR^3}}\,t\right) + B\sin\left(\sqrt{\frac{GM}{mR^3}}\,t\right)$$

(ω under each square root)

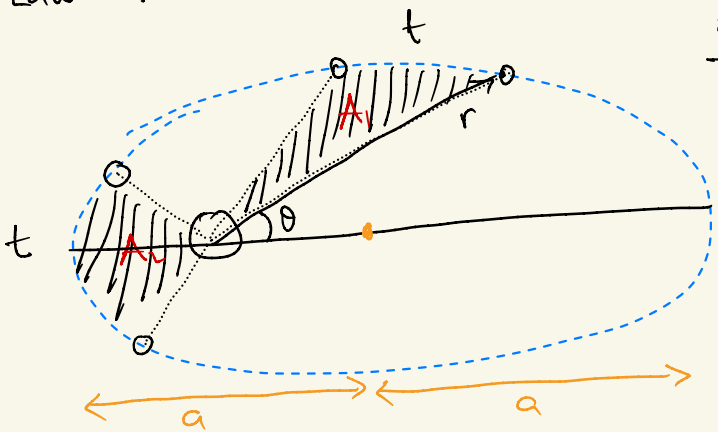
Kepler's Three laws (1571 - 1630)

① The law of orbits: All planets move in elliptical orbits.



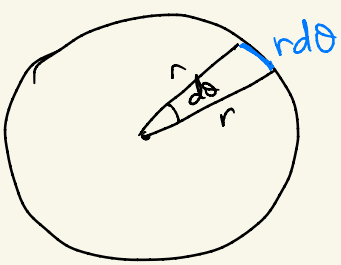
② Law of areas

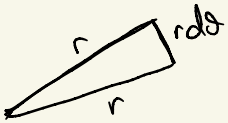
$A_1 = A_2$



$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{1}{2} r^2 \omega$$

when $d\theta$ is infinitesimally small, this is basically a triangle





$$\frac{(r)\,r d\theta}{2} = dA$$

③ Law of Periods:

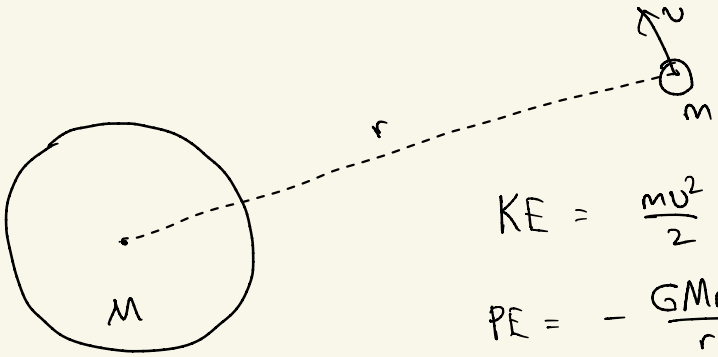
$$\frac{GM\cancel{m}}{r^2} = m a = \cancel{m}\omega^2 r$$

$T^2 : R^3$ relationship

$$\frac{GM}{r^3} = \left(\frac{2\pi}{T}\right)^2 = \frac{4\pi^2}{T^2}$$

$$T(r) = \sqrt{\frac{4\pi^2 r^3}{GM}} \quad \leftarrow \quad \boxed{\frac{T^2}{r^3} = \frac{4\pi^2}{GM} = \text{const.}}$$

Total Energy of Gravitational system of planets



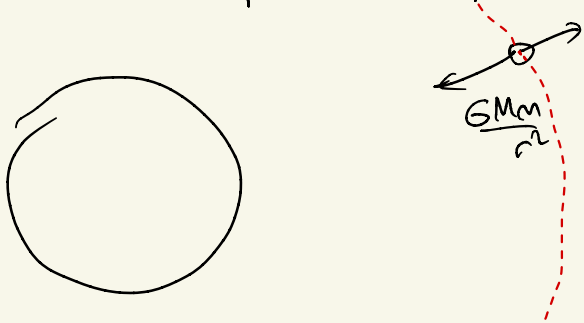
$$KE = \frac{mv^2}{2}$$

$$PE = -\frac{GMm}{r}$$

$$E_{mechanical} = KE + PE$$

$$= \frac{mv^2}{2} - \frac{GMm}{r}$$

Ми знаємо с равновесие орбити $\frac{mv^2}{r} = \frac{GMm}{r^2}$



$$\frac{mv^2}{2} = \frac{GMm}{2r}$$

$$E_{mech.} = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r} \star$$

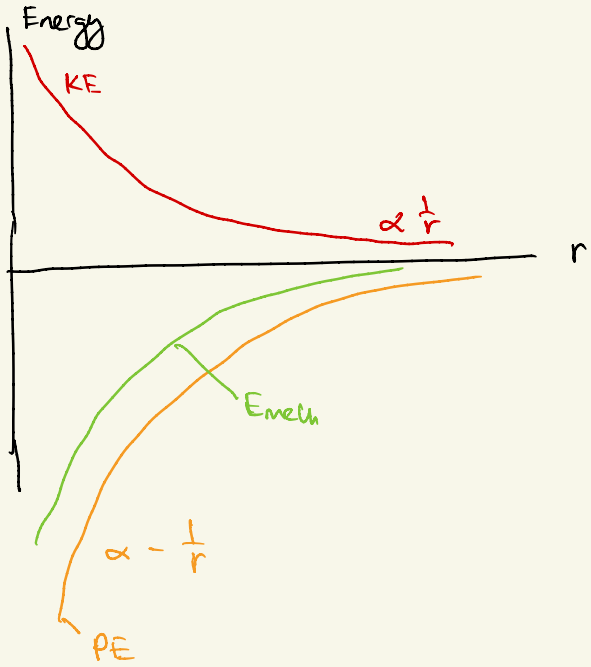
Note: Negative total energy is usually called "Binding Energy" in other areas of physics: atomic Nuclear Quantum

Here, a: semi major axis

$a = \frac{\text{apogee} + \text{perigee}}{2}$

$E_{mech} = -\frac{GMm}{2a}$ (in elliptical orbits)

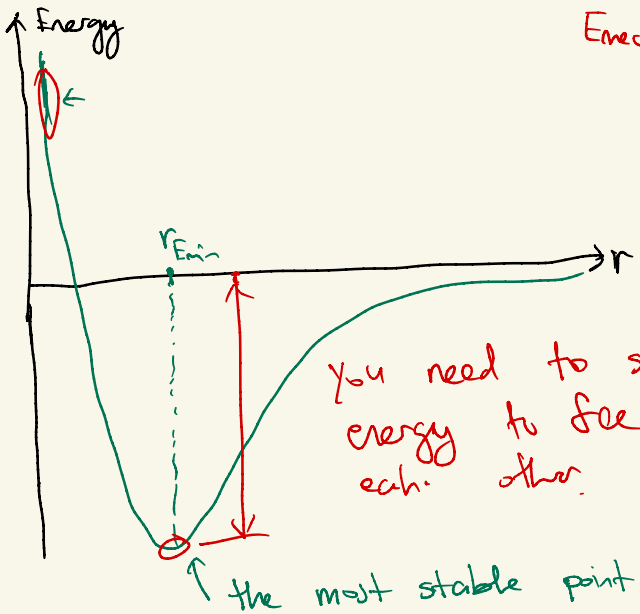




$KE = \frac{mv^2}{2} = \frac{GMm}{2r}$

$PE = -\frac{GMm}{r}$

If you remember, "Lennard Jones Potential" in molecules



$E_{mech} = 0$: free particles

you need to supply this much energy to free particles from each other.

↑ the most stable point

Conservative forces and Conservative potentials

10 Figure 13-28 shows six paths by which a rocket orbiting a moon might move from point *a* to point *b*. Rank the paths according to (a) the corresponding change in the gravitational potential energy of the rocket–moon system and (b) the net work done on the rocket by the gravitational force from the moon, greatest first.

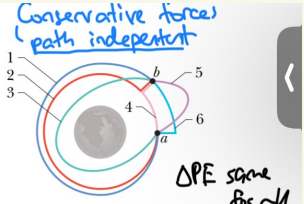


Fig. 13-28 Question 10.

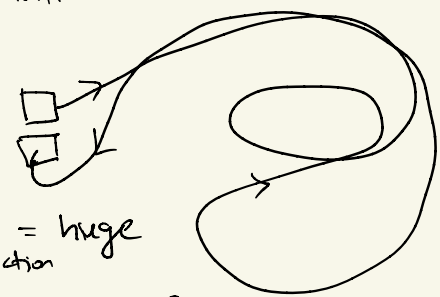
11 Figure 13-29 shows three uni-

=

Path independent. Only final and initial point matter.

Friction force ≠ conservative

Gravitational force = conservative



$W_{done\ by\ friction}$ = huge

$W_{done\ by\ gravity}$ = ΔPE = 0

(a) 1 = 2 = 3 = 4 = 5 = 6

(b) 1 = 2 = 3 = 4 = 5 = 6

•••**16** In Fig. 13-39, a particle of mass $m_1 = 0.67$ kg is a distance $d = 23$ cm from one end of a uniform rod with length $L = 3.0$ m and mass $M = 5.0$ kg. What is the magnitude of the gravitational force $\vec{F}$ on the particle from the rod?

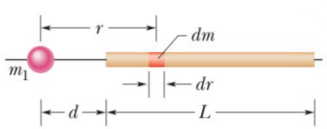


Fig. 13-39 Problem 16.

$$dF = -\frac{Gm_1\,dm}{r^2}$$

$$F = \int dF = -\int_d^{L+d} \frac{Gm_1\,\lambda\,dr}{r^2} = -Gm_1\left(\frac{M}{L}\right)\left[\int_d^{L+d}\frac{dr}{r^2} = -\frac{1}{r}\Big|_d^{L+d}\right]$$

$$= -\frac{Gm_1M}{L}\left(\frac{1}{d} - \frac{1}{L+d}\right)$$

••13 Figure 13-37 shows a spherical hollow inside a lead sphere of radius $R = 4.00$ cm; the surface of the hollow passes through the center of the sphere and "touches" the right side of the sphere. The mass of the sphere before hollowing was $M = 2.95$ kg. With what gravitational force does the hollowed-out lead sphere attract a small sphere of mass $m = 0.431$ kg that lies at a distance $d = 9.00$ cm from the center of the lead sphere, on the straight line connecting the centers of the spheres and of the hollow?

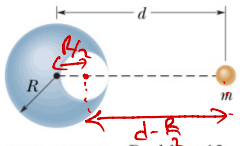
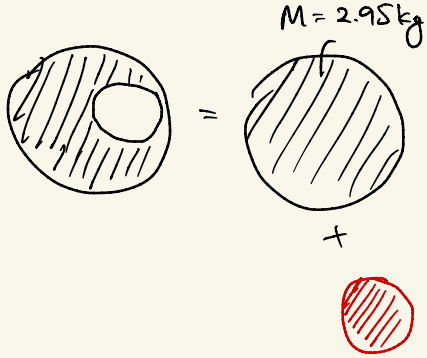


Fig. 13-37 Problem 13.



$$F_{big\,sphere} = -\frac{GMm}{d^2}$$

$$F_{positive} = +\frac{GM_pm}{\left(d-\frac{R}{2}\right)^2}$$

$$F_{total} = F_{big\,sphere} + F_{positive} = -\frac{GMm}{d^2} + \frac{GM_pm}{\left(d-\frac{R}{2}\right)^2}$$

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3} \Rightarrow M = \rho\cdot\frac{4}{3}\pi R^3$$

$$M_p = \rho\cdot\frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{M}{8}.$$

$$F_{tot.} = -\frac{GMm}{d^2} + \frac{GMm}{8\left(d-\frac{R}{2}\right)^2}.$$

$$F_G = -\frac{GM_1M}{r^2}$$ ★

sec. 13-5 Gravitation Inside Earth

•24 Two concentric spherical shells with uniformly distributed masses M_1 and M_2 are situated as shown in Fig. 13-40. Find the magnitude of the net gravitational force on a particle of mass m, due to the shells, when the particle is located at radial distance (a) a, (b) b, and (c) c.

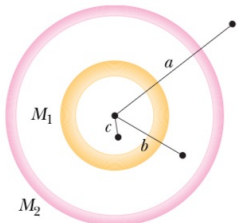


Fig. 13-40 Problem 24.

••25 A solid uniform sphere has a mass of 1.0×10^4 kg and a radius of 1.0 m. What is the magnitude of the

(a) $F = -\frac{G(M_1+M_2)m}{a^2}$

(b) $F = -\frac{GM_1m}{b^2}$

(c) $F = 0$.

Binary Star : IPhO 2011

It was a bit detailed version of this involving reduced mass $\mu = \frac{m_1m_2}{m_1+m_2}$

••54 *Hunting a black hole.* Observations of the light from a certain star indicate that it is part of a binary (two-star) system. This visible star has orbital speed $v = 270$ km/s, orbital period $T = 1.70$ days, and approximate mass $m_1 = 6M_s$, where M_s is the Sun's mass, 1.99×10^{30} kg. Assume that the visible star and its companion star, which is dark and unseen, are both in circular orbits (Fig. 13-46). What multiple of M_s gives the approximate mass m_2 of the dark star?

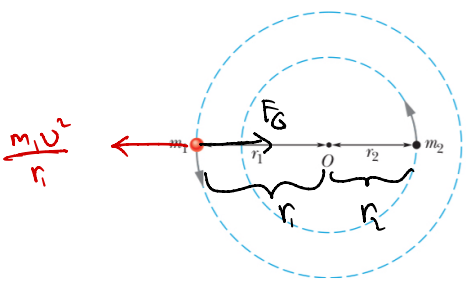


Fig. 13-46 Problem 54.

$\frac{m_1v^2}{r_1}$

$$F_G = -\frac{Gm_1m_2}{(r_1+r_2)^2}$$

Equation for 2nd star

Solution:

$$\frac{m_1v_1^2}{r_1} = \frac{Gm_1m_2}{(r_1+r_2)^2}$$

$$\frac{Gm_2m_1}{(r_1+r_2)^2} = \frac{m_2v^2}{r_2}$$

$$\boxed{\frac{(r_1+r_2)^2}{r_1} = \frac{Gm_2}{v_1^2}}$$

$$\boxed{\frac{Gm_1}{v_2^2} = \frac{(r_1+r_2)^2}{r_2}}$$

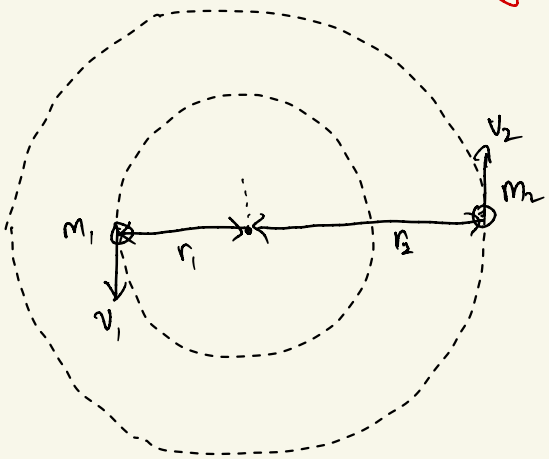
$$\frac{(r_2+r_1)^2}{r_1} = \frac{Gm_2}{\left(\frac{2\pi r_1}{T}\right)^2}$$

$$\frac{(r_2+r_1)^2}{r_2} = \frac{Gm_1}{\left(\frac{2\pi r_2}{T}\right)^2}$$

$$\frac{(r_1+r_2)^2 r_1^{\cancel{x}}}{\cancel{r_1}} = \frac{GM_2T^2}{4\pi^2}$$

$$\frac{(r_1+r_2)^2 r_2^{\cancel{x}}}{\cancel{r_2}} = \frac{Gm_1T^2}{4\pi^2}$$

⇒ Neat Solution for Problem 54 (Note: solving Cubic eq'n is not part of usual olympic curriculum. At least, I know so).



$$\frac{G m_1 m_2}{(r_1+r_2)^2} = \frac{m_1 v_1^2}{r_1}$$

$$(r_1+r_2)^2 = \frac{G m_2 r_1}{v_1^2}$$

$$\boxed{r_1+r_2 = \sqrt{\frac{G m_2 r_1^2}{v_1^2}}}$$

Also, compute r_1 from center of mass

$$\frac{m_1 \cdot 0 + m_2 (r_1+r_2)}{m_1+m_2} = r_1$$

$$\boxed{r_1+r_2 = \frac{m_1+m_2}{m_2} r_1}$$

Equate two eqns:

$$r_1+r_2 = r_1+r_2$$

$$\frac{m_1+m_2}{m_2} r_1 = \sqrt{\frac{G m_2 r_1}{v_1^2}}$$

$$\left(\frac{m_1}{m_2}+1\right)^2 r_1^2 = \frac{G m_2 r_1}{v_1^2}$$

$$(m_1+m_2)^2 = \frac{G}{v_1^2 r_1} m_2^3 \longrightarrow$$

$$\frac{v_1}{T} = \frac{2\pi r_1}{T} = v_1$$

$$r_1 = \frac{v_1 T}{2\pi}$$

$$\boxed{\frac{m_2^3}{(m_1+m_2)^2} = \frac{v_1^2 r_1}{G} = \frac{v_1^3 T}{2\pi G}}$$

$$\frac{m_2^3}{(m_1+m_2)^2} = \frac{v_1^3 T}{2\pi G} = \frac{(270000\ m\cdot s^{-1})^3\left[(1.70\ \cancel{days})(24\ \cancel{hr}/\cancel{day})(3600\ s/\cancel{hr})\right]}{(2\pi)(6.67\times10^{-11}\ kg^{-1}\cdot m^3\cdot s^{-2})}$$

$$= 6.90\times10^{30}\ kg$$

$$= 3.467 \times (1.99\times10^{30}\ kg)$$

$$= 3.467\ M_S$$

We're given that $m_1 = 6M_S$, we can write $m_2 = \alpha M_S$

$$\frac{m_2^3}{(m_1+m_2)^2} = \frac{(\alpha M_S)^3}{(6M_S+\alpha M_S)^2} = 3.467\ M_S$$

$$\frac{\alpha^3}{(6+\alpha)^2} = 3.467$$

$$\boxed{\frac{\alpha^3}{(6+\alpha)^2} - 3.467 = 0} \Rightarrow \alpha^3 - (6+\alpha)^2 3.467 = 0$$

$$\alpha^3 - (6^2 + 2\cdot6\cdot\alpha + \alpha^2)\,3.467 = 0$$

$$\alpha^3 - 36\cdot3.467 - 12\alpha\cdot3.467 - 3.467\alpha^2 = 0$$

$$\alpha^3 - 3.467\alpha^2 - 41.6\alpha - 36 = 0$$ ⟩ Cubic eq'n

$$ax^3 + bx^2 + cx + d = 0$$

with $a = 1$, $b = -3.467$, $c = -41.6$, $d = -36$

α has 3 roots: $\alpha_1 = -4.28$

$\alpha_2 = -0.96$

$\boxed{\alpha_3 = 8.71}$ → being only real solution because mass can't be negative. $\boxed{m_2 = \alpha M_S = 8.71 M_S}$

•••58 The presence of an unseen planet orbiting a distant star can sometimes be inferred from the motion of the star as we see it. As the star and planet orbit the center of mass of the star–planet system, the star moves toward and away from us with what is called the *line of sight velocity*, a motion that can be detected. Figure 13-48 shows a graph of the line of sight velocity versus time for the star 14 Herculis. The star's mass is believed to be 0.90 of the mass of our Sun. Assume that only one planet orbits the star and that our view is along the plane of the orbit. Then approximate (a) the planet's mass in terms of Jupiter's mass m_J and (b) the planet's orbital radius in terms of Earth's orbital radius r_E.

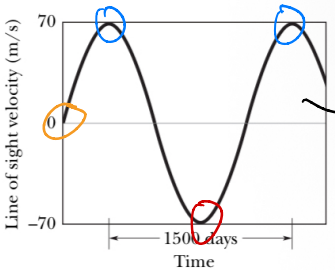


Fig. 13-48 Problem 58.

Binary star

$m_s = 0.9\, M_{sun}$

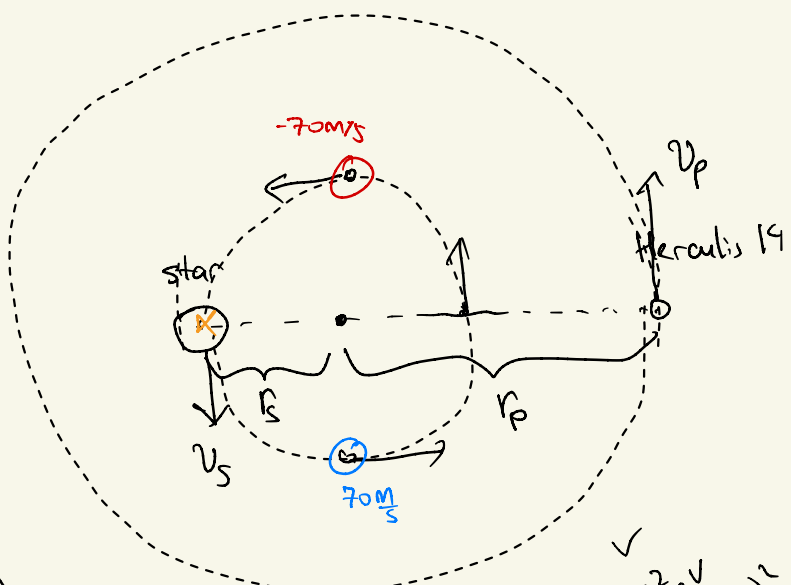
$v_s = \frac{2\pi r_s}{T} \Rightarrow r_s = \frac{v_s T}{2\pi}$

v_s (from graph) $= 70\ \frac{m}{s}$

T (from graph) $= 1500\ days \times \frac{24h}{day} \times \frac{3600s}{h}$

$r_s = \frac{v_s T}{2\pi} = \frac{(70\,ms^{-1})(1.296\times10^8\,s)}{2\pi}$

$= 1.444\times10^9\ m$



Earth ≡

✓ ✗ ✓ ✓

$\frac{GM_s M_p}{(r_s+r_p)^2} = \frac{M_s v_s^2}{r_s}$

$\frac{GM_s M_p}{(r_s+r_p)^2} = \frac{M_p v_p^2}{r_p}$

$\frac{M_s \cdot 0 + M_p(r_s+r_p)}{(M_s+M_p)} = r_s$

$\frac{GM_p}{(r_s+r_p)^2} r_s = v_s^2 \rightarrow M_p = \frac{v_s^2 (r_s+r_p)^2}{G r_s}$ ✓ ✓

$\frac{GM_s}{(r_s+r_p)^2} r_p = v_p^2 = \left(\frac{2\pi r_p}{T}\right)^2$

$$\frac{M_p(r_s+r_p)}{M_s+M_p} = r_s$$

$$(r_s+r_p) = \frac{M_s+M_p}{M_p} r_s$$

$$M_p = \frac{v_s^2 (r_s+r_p)^2}{G r_s} = \frac{v_s^2 \left(\frac{M_s+M_p}{M_p}\right)^2 r_s^2}{G r_s}$$

$$\frac{GM_p}{r_s} = v_s^2 \left(\frac{M_s}{M_p}+1\right)^2$$

$$\frac{GM_s}{r_s v_s^2} = \left(\frac{M_s}{M_p}+1\right)^2 \frac{M_s}{M_p}$$

$$\frac{M_s}{M_p} = x, \qquad \frac{GM_s}{r_s v_s^2} = (x^2+2x+1)\,x$$

$$x^3+2x^2+x-\frac{GM_s}{r_s v_s^2} = 0$$

Recall $ax^3+bx^2+cx+d=0$ Cubic eqn with $a=1$, $b=2$, $c=1$, $d=-\frac{GM_s}{r_s v_s^2}$

$$d = -\frac{(6.67\times10^{-11}\,kg^{-1}m^3s^{-2})(0.9\times1.99\times10^{30}\,kg)}{(1.444\times10^{9}\,m)(70\,m/s)^2} = -1.69\times10^{7}$$

Our Cubic eqn has 3 roots: (see attached Cubic eqn solver)

$x_1 = -128.98 + 222.2i$
$x_2 = -128.98 - 222.2i$
} complex values

$x_3 = 255.96$ real

So the only possible solution is $x = \frac{M_{star}}{M_p} = 255.96$ because $\frac{M_{star}}{M_p}$ = can't be negative, let alone a complex value!

$$M_p = \frac{M_{star}}{255.96} = \frac{0.9\,M_{sun}}{255.96}$$

$$= \frac{0.90 \times 1.99 \times 10^{30}\ kg}{255.96}$$

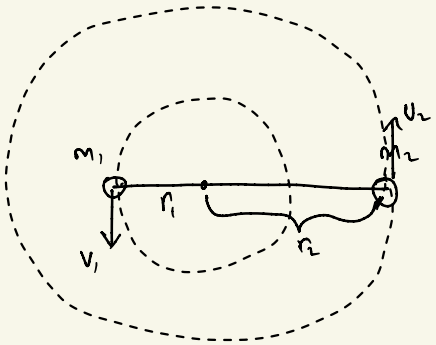
$$= 6.994 \times 10^{27}\ kg$$

$$= 3.68\ M_{Jupiter}$$

Note that $M_{sun} = 1.99 \times 10^{30}\ kg$

$M_{Jupiter} = 1.898 \times 10^{27}\ kg$

Things to know for Binary stars:



① $F_G = F_{centrifugal}$

$$\frac{GM_1m_2}{(r_1+r_2)^2} = \frac{m_1v_1^2}{r_1}$$

$$\frac{Gm_1m_2}{(r_1+r_2)^2} = \frac{m_2v_2^2}{r_2}$$

② Center of mass

$$\frac{m_1 \cdot 0 + m_2 \cdot (r_1+r_2)}{m_1+m_2} = r_1$$

HW for Ch13

Q: 5

Problems: 27, 33, 68, 71, 79, 80, 85, 84,